

SSc-MIM

A curated, SBGN-compliant Molecular Interaction Map of
skin fibrosis in diffuse cutaneous systemic sclerosis

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github.com/Nurtal/IDT_SSc_map

Systemic sclerosis lacks a mechanistic map

The disease

Systemic sclerosis (scleroderma) — autoimmune fibro-inflammatory disease: progressive skin & organ fibrosis, vasculopathy, immune dysregulation.

Highest mortality of any rheumatic disease — and no disease-modifying therapy is approved.

The gap

Curated disease maps already exist for rheumatoid arthritis (RA-map), Sjögren, COVID-19, Parkinson...

None exists for SSc. Its core molecular circuits are scattered across the literature, never assembled into one mechanistic model.

Build the first SSc disease map, focused on dcSSc skin fibrosis.

Standards-based, fully sourced, and citable (GitHub + Zenodo DOI).

Layered with real single-cell patient data so it is not just a diagram but an analysis substrate.

What is a Molecular Interaction Map?

A computable, standards-based diagram of the molecular events driving a disease — species (genes, proteins, complexes, phenotypes) connected by mechanistic reactions.

SBGN

Systems Biology Graphical Notation — an unambiguous visual grammar for biology

CellDesigner

The editor the map is built in; exports standard SBML (machine-readable)

MI2CAST

Annotation standard: every reaction carries its evidence, source PMID & ECO code

Reproducible

Public repo, scripted pipeline, Docker, continuous-integration validation

The SSc-MIM at a glance

568

molecular species

308

mechanistic reactions

20

cell & tissue compartments

4

biological modules

133hand-curated SSc-specific
reactions**~360**PubMed references mined &
filled**197**

patients across 4 single-cell datasets overlaid

Built in CellDesigner, validated as SBML (0 errors in CI), annotated to MI2CAST, and released openly. The map is organised so that every disease-relevant entity flows toward a measurable disease phenotype.

Four biologically coherent modules

M1 — Type-I interferon

cGAS-STING, RIG-I/MAVS, TLR3/7/9 → IRF3/7 → ISG signature.
The innate-immune ignition of SSc.

M2 — TGF- β & fibrosis

Latent TGF- β activation → SMAD2/3 → fibroblast-to-myofibroblast transition, collagen & ECM. The fibrotic core.

M3 — EndoMT & vasculopathy

Endothelin, Notch, NO/sGC, HIF → endothelial-to-mesenchymal transition & vascular remodelling.

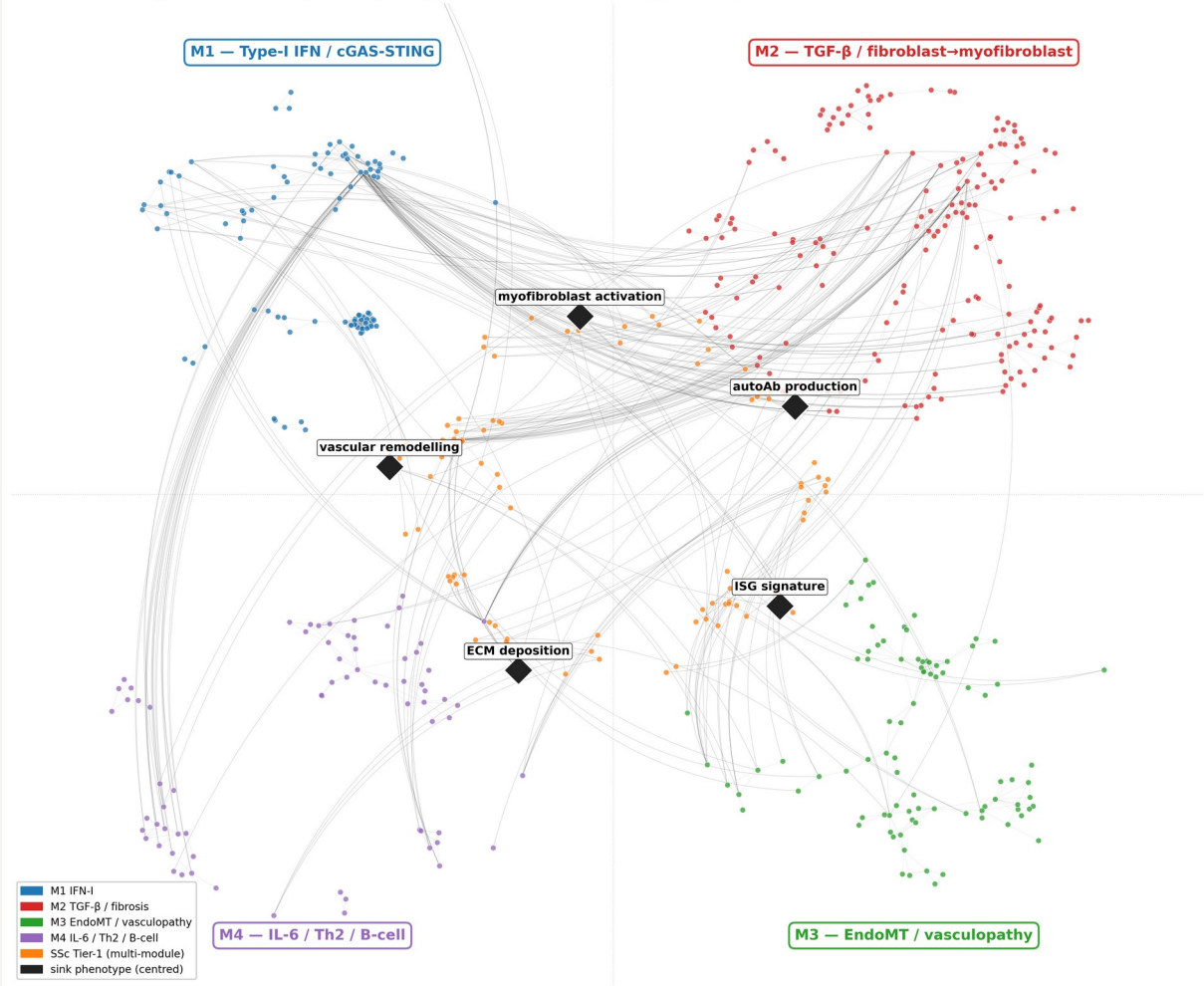
M4 — IL-6 / Th2 / B cells

IL-6/STAT3, IL-4/IL-13/STAT6, BAFF-BCMA, autoantibodies. The adaptive-immune & cytokine layer.

Plus a crosstalk layer wiring the modules together (e.g. IL-6 → SMAD3, IFN-I → fibroblast).

One integrated, modular network

Figure 1 — SSc-MIM global view (quadrant layout)
526 species · 1488 edges · modules in fixed positions · phenotype sinks centred · inter-module edges shown as curved arcs



Each quadrant is one module; nodes are molecular species, edges are mechanistic reactions.

Hubs that emerge from network analysis match known SSc biology: TGF- β :receptor, SMAD3–SMAD4, ISG signature, PDGFR.

The same SBML file drives the figure, the analysis, and the online viewer.

From molecules to disease phenotypes

The map is built around four phenotype “sinks” — the biological outputs of the disease. Modelling rule: every disease-relevant entity must reach a sink in ≤ 6 steps (verified: 0 violations).

**Myofibroblast
activation**

**ECM / collagen
deposition**

**Vascular
remodelling**

**ISG / interferon
signature**

It turns a static drawing into a testable model: a perturbation anywhere has a traceable path to a clinical readout.

It makes the scope explicit and reviewable — exactly what we want expert biologists to challenge.

Every reaction is sourced and graded

Tiered citation policy (GO-aligned)

Each reaction carries a primary PubMed ID and an ECO evidence code — experimental evidence ranked above inference.

Provenance is tracked

Every edge records where it came from: Reactome import, hand curation, or literature mining — fully auditable.

No fabricated citations — ever

A built-in grounding gate rejects any claim without a real, verifiable source (a deliberate negative-control proves it works).

~360 references mined and auto-filled (title, authors, journal) via NCBI E-utils.

133 SSc-specific reactions curated from a systematic 77-paper worklist + co-author full-text PDFs.

MI2CAST-compliant annotation throughout.

Layered with real patient single-cell data

4 datasets • 197 patients

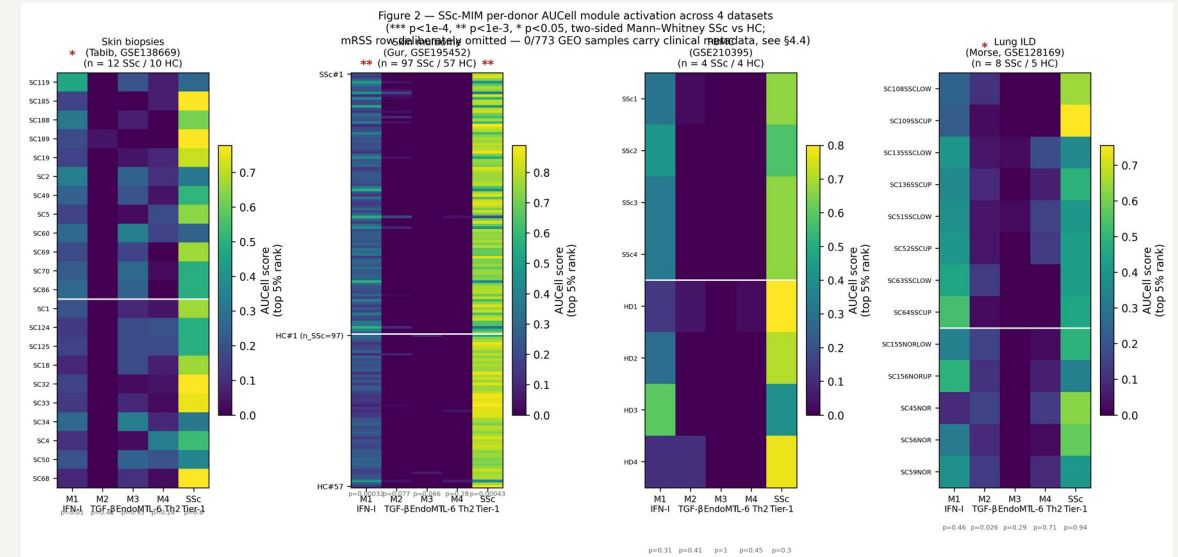
Skin (Tabib, Gur multiome), PBMC and lung — SSc vs healthy controls.

81% of the map is measurable

After mixed-effects differential expression, 81% of detectable map species are observed in the data.

The biology holds up

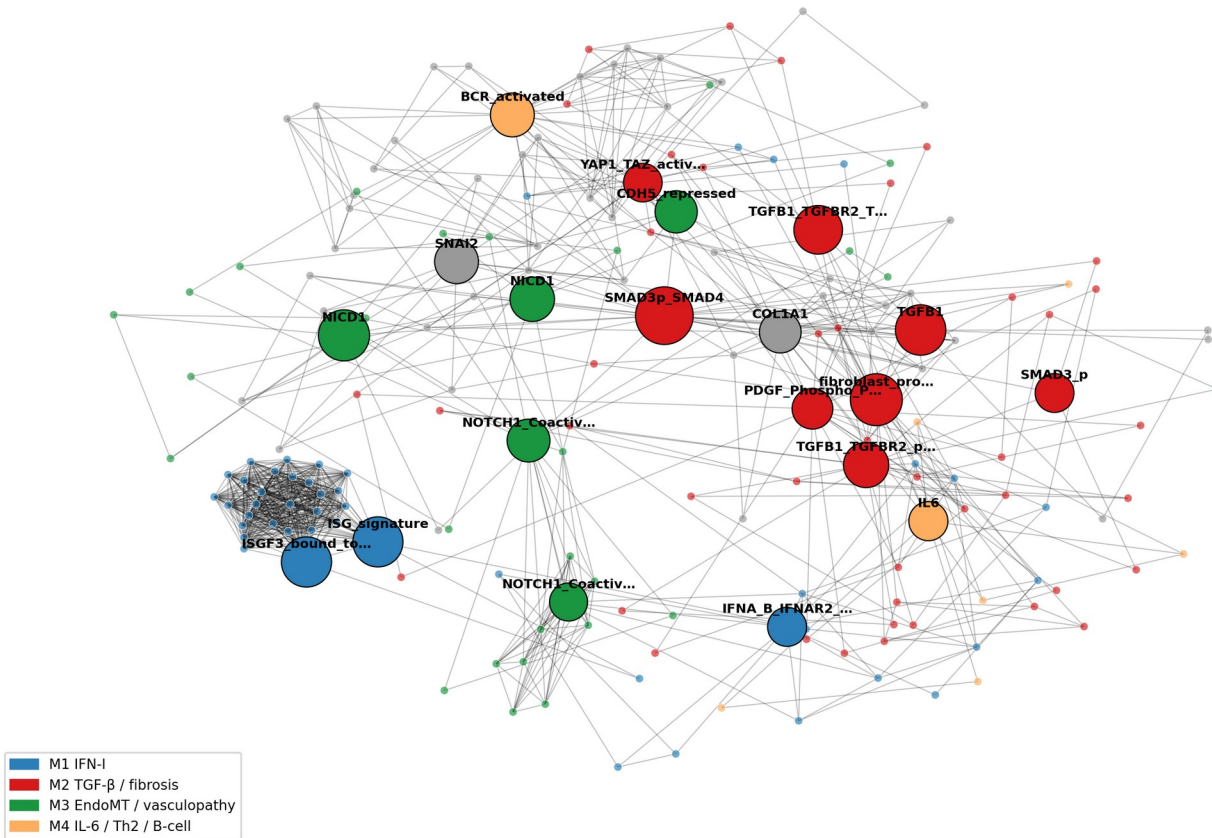
Per-donor module scoring (AUCell) shows the M1 interferon signature significantly elevated in SSc skin ($p = 3.2 \times 10^{-4}$).



AUCell module scores, SSc vs HC, per donor (Fig. 2).

A substrate for drug prioritisation

F3 (preview) — top-20 hub subnetwork in SSc-MIM
size $\propto$ hub_score · colour = source module



Network hubs cross-referenced with DGIdb drug-target data.

Recovers targets already in SSc trials — tocilizumab (IL-6), nintedanib, romilkimab (IL-13), pamrevlumab (CTGF/CCN2), bosentan (endothelin).

Drug table recalibrated against real SSc clinical-trial outcomes.

Shows the map can rank and rationalise repurposing candidates.

Independent verification of every interaction

143

interactions queued for expert review

128

passed automated AI verification

28

wrong source PMIDs caught & fixed

0

fabricated citations

Contradiction detector

Flags any gene pair given opposite signs by different sources, for human arbitration.

AI reviewer pass

Each interaction was checked against its deciding sentence and supporting/contrary abstracts — verdict: validate / revise / caution.

Citation-integrity sweep (this week)

Detected 28 reactions whose cited PMID pointed to an unrelated paper (biology canonical, reference a data-entry slip — e.g. a TGF- β reaction cited to a rubella study). Each replacement PMID was re-verified against live PubMed before applying.

An app to review the map, one interaction at a time

A swipe deck for curation

Offline, self-contained web app — one card per interaction, accept / reject / annotate.

Everything you need to decide, on the card

Regulator → target, mechanism, SSc relevance, the verbatim deciding sentence from the source article, and PubMed/DOI links.

Context before you judge

A literature dossier lists candidate supporting and “possibly contrary” references; an in-browser module map shows the neighbourhood.

Advisory AI verdict

Each card shows the AI call + rationale — you can filter to triage, but the human reviewer always decides.

No install, no server — open one HTML file.

Decisions saved locally; export to CSV/JSON.

Re-includes even discarded edges, so you keep full control.

Demo right after this.

Three things from expert biologists

1

Validate the scope

Do the 4 modules + phenotype sinks capture dcSSc skin fibrosis? What is missing or over-reaching?

2

Curate the entities

Tier-1 SSc species: what to add, remove, or promote — with PMIDs where you have them.

3

Adjudicate interactions

Use the review app on the 143 interactions: confirm, reject, or flag for revision.

Where this is going

An open, citable SSc resource — and your review is the gate

v1.0 release: GitHub + Zenodo DOI (citable).

Resource paper (npj Systems Biology & Applications) — revision drafted.

Expert sign-off on scope & interactions is the last binding step.

Stretch: interactive MINERVA web deployment.

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github.com/Nurtal/IDT_SSc_map

Open the review app now for a live walkthrough.